

Earth Science Final Project: PFOAs and PFOS Concentration in Relation to the Ocean

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Introduction

This PFOAs and PFOS in Massachusetts pose a significant risk to people's health. Every day, millions of people ingest forever chemicals in their water. They are almost unavoidable and lead to increased risk of many health conditions. Some include increased risk of obesity, cancer, and infertility. These "forever chemicals" get into our water in different ways. The chemicals often come from water resistant products, fire extinguishers, and released chemicals.

Purpose

The importance of understanding the spread of PFAs cannot be understated. With the health risks of our own drinking water, limiting spread is pertinent. We often see heavy concentrations in densely populated areas. In this project, I was interested to see if there is a way in which the spread could be observed to impact some areas more than others in their relation to their distance from the ocean. In finding a correlation like this, we could further understand the spread of PFAs and try different relevant approaches to limit it.

Method

In this study, I wanted to analyze the concentration of PFOAs and PFOS in relation to the distance from the ocean. Using the ewg interactive map, I looked for an area which was heavily concentrated with points over the reportable limit. I found that Massachusetts had a good balance of measurements inland and by the ocean. This offered a good site to observe.

Data

Although these chemicals have differing names, PFOAs and PFOS are both under the category of PFAs. There were several chemicals in the EPA data which were considered, but these two had the most relevant observations which would yield the best results. The intention of this project wasn't to force a correlation and there would be a risk doing that with such limited observations as found in different chemicals.

To answer this question of whether PFAs concentrations are higher (or lower) depending on proximity to the ocean, getting a good dataset was pertinent. Using the EPA's website with information on detected concentrations in drinking water, I was able to make a chart with the relevant chemicals and site information.

Sorting

The EPA website allows anyone to select which parameters they want in the data. As there are millions of observations, it's impossible to generate a table of all of the data. For the parameters in this project, the only state which was observed was Massachusetts. The contaminants were set to PFOAs and PFOS only.

After generating this table and importing it into R, there were limitations set to the contaminants. The most common result was "< MRL". This made the data impossible to work with as it was essentially saying 0 although there were undoubtedly values there which just could not be detected. Removing the "< MRL" values allowed for work with real values, as there were too many missing to properly infer what their values would be. For every ten locations there was one with measurement of PFOAs or PFOS above this limit.

Starting to work with the data, PFOAs and PFOS also had to be split. With this new subset of data with only numbers, the data was further split by the two types of contaminants. This completely separated the data in the ways it would normally need to be used in data analysis.

For this project in particular, location was very important given it is a comparison of the concentration at different distances from the ocean. Each site that the data was taken is listed in the sheet. Because there is no metric of distance from any point with all these locations, getting the distance of each from the ocean would have been a long process. Copilot was able to analyze the data and sort the distances into groups of "close", "medium", and "far" describing their distances from the ocean. Please see the "AI Usage" section for more information.

Techniques

With the data now organized by contaminant type and distance from the ocean, I could begin to work with the data. I started by creating different visuals to better understand the data, and then moved to analyzation techniques like t-tests.

AI Usage

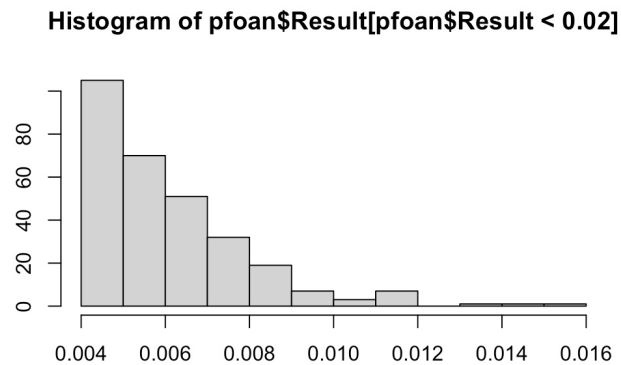
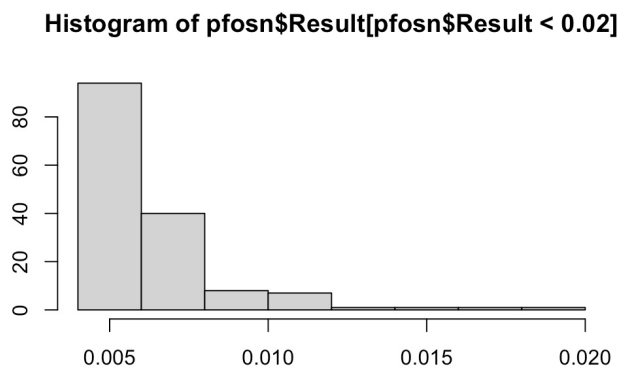
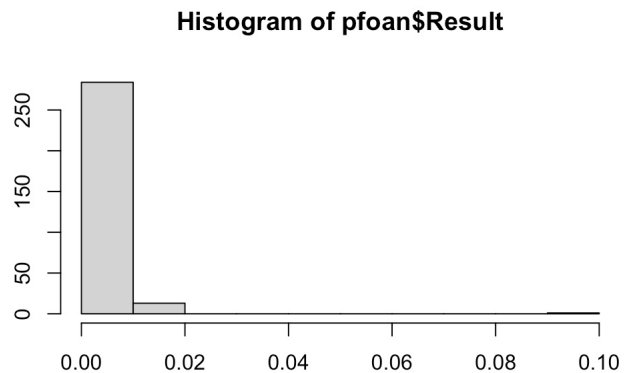
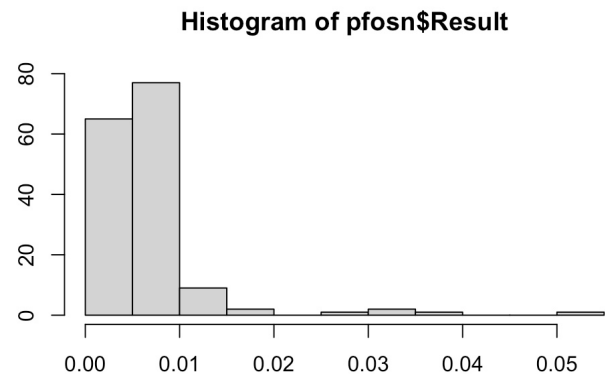
I asked copilot for help with sorting the distance data as there were many locations and I was worried I would make many errors manually sorting this. I said to sort each by 0-10 miles from the ocean, 10-20 miles, and 20 or more. Copilot gave me lists of each which I then copied into R as "close", "med", and "far". The lists I got from copilot are in the github table section. Looking up each site and mapping the distance from the ocean would've taken hours if not days to do manually and didn't take anything away from the problem solving of this project so I think it was a reasonable use of AI.

Results

This section will discuss the findings, interpret the figures, and analyze the table results. All figures have the PFOS figure on the left and PFOA figure on the right.

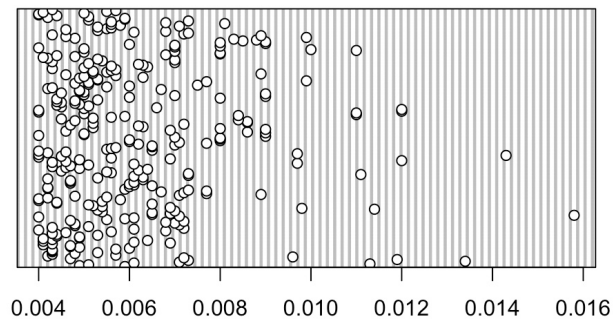
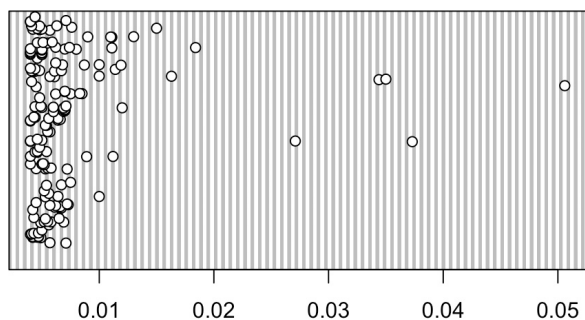
Figures

Histograms



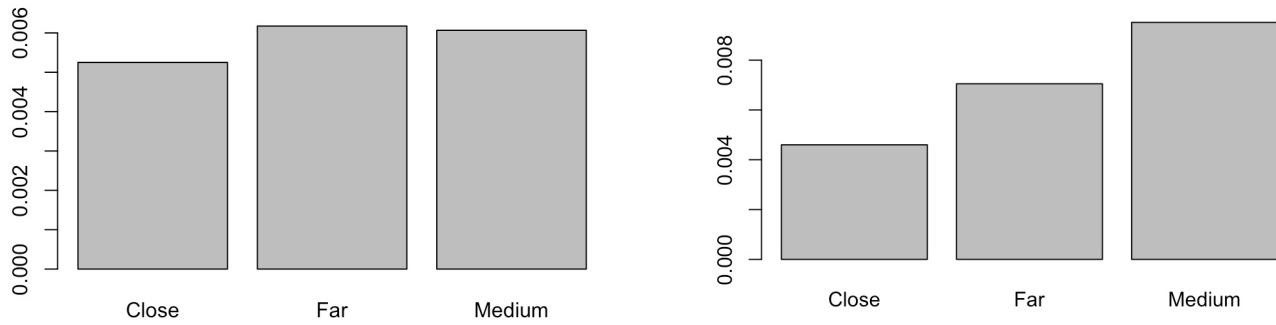
The histograms came out with little data initially. Seeing the range from 0 to 0.05 on PFOS and 0 to 0.10 on PFOAs, it was clear a more detailed view was important. Confining the ranges to 0.02 made it so that the general distribution trend of these contaminants were much more visible. The histograms were not intended to show anything regarding the distance from the ocean. They're simply here to give an idea of the concentration at different levels.

Dot Chart



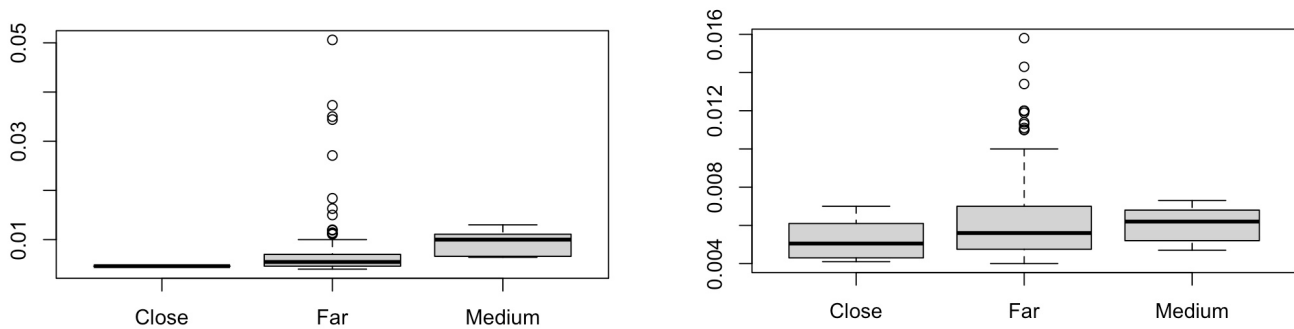
The dot charts are similar to the histograms, but they give a more detailed view of the concentrations. We can see very clearly how many are in each region giving a very good visualization of the way that the concentration stays at a relatively low value.

Bar Chart



The bar charts now take location groups into account. Here we can see the average amount in the close, far, and medium groups of each contaminant side by side. This was the first indication that there was no observable pattern in these concentrations with relation to the ocean. The PFOA chart does look at though there may be a pattern ordered in the way it is, but there is no reason there would be the jump from close, far, and then medium if there was this relationship correlation. The PFO chart could have more of a relation given the close is smaller than the medium which is smaller than the far, but given the dot chart is likely due to the several outliers in the far section skewing the data. That could be explained by several things, but with such a drastic spike most likely places releasing waste which contaminants the drinking water.

Box Plots



As we see again this provides further evidence that the outliers are skewing the means in the data for the far section which is the only part of the data which could indicate some upward trend in distance from the water and concentration of PFOS. The tests could give further information on whether there is any correlation between the distances, as well as look for some sort of correlation between the PFOS and PFOAs if there is some sort of abnormal trend.

Tables

ANOVA

Group	Df	Sum Sq	Mean Sq	F value	p-value
Distance	2	6.70E-06	3.357E-06	0.881	0.415
Residuals	294	1.12E-03	3.810E-06	—	—

Table 1: PFOA

Group	Df	Sum Sq	Mean Sq	F value	p-value
Distance	2	4.80E-05	2.394E-05	0.651	0.523
Residuals	155	5.699E-03	3.676E-05	—	—

Table 2: PFOS

The ANOVA test was good for this data because it considers the distances in each, allowing us to see how good the correlation is between the distance factor independent of the other. Both outcomes were bad with very high p-values making a real connection very unlikely. We would want to see a p-value closer to 0.05, but both are almost ten times the value which constitutes a good p-value.

Variance

Contaminant	Variance
PFOA	3.81×10^{-6}
PFOS	3.66×10^{-5}

PFOS had a much higher variance than PFOAs. With an extra factor of ten, it is very unlikely that the PFOS data has a similar relationship to that of the PFOA data, with the PFOS data being much more sporadic than the PFOA data. This is easily observable in the dot charts as although the PFOA data is stretched across more points, a larger portion of PFOS data is stretched further across proportionally larger points. This is good to know for things comparing the PFOS and PFOA data as understanding one has higher variance can make the data less reliable.

Standard Deviation

Contaminant	Standard Deviation
PFOA	1.95×10^{-3}
PFOS	6.05×10^{-3}

This is the gold standard of variance tests. This is a much more straightforward version of the previous calculation. Given the variance we saw that the variance of the PFOAs were significantly lower than PFOS. The range of points in three standard deviations of PFOAs, or over 99 percent of data in a normal distribution, is just in one standard deviation of the PFOS. With such a large difference, the way in which the PFOS distribution acts is clearly very different from that in which the PFOAs data does.

T-test

Contaminant	t-statistic	p-value
PFOS	$t(149) = -4.87$	2.84×10^{-6}
PFOA	$t(8.31) = -2.23$	0.055

The t-test was done on just the close and far distances. This test made sense given that these are the two extremes. If there was data skewed in the middle making a relationship unobservable, this t-test would make up for it. The p-value in the PFOS actually was very small. This could indicate a difference in values and is a very good p-value for data like this. The PFOA value on the other hand was a very large p-value. This means there is likely no real association between the PFOA concentration and it's distance from the ocean.

Regression

Variable	Estimate	Std. Error	t value	p-value
Intercept	0.0052500	0.0006901	7.608	3.8×10^{-13}
Far	0.0009250	0.0006999	1.322	0.187
Medium	0.0008167	0.0009484	0.861	0.390

$$R^2 = 0.00596, \text{ Adjusted } R^2 = -0.00080, F(2, 294) = 0.881, p = 0.415$$

The regression again shows that there is likely no real correlation between these distance groups from the ocean and the concentration of PFAs. With a very low R^2 value it seems to have no real correlation. This test was good to see if there was a way to find predict values and given the poor outcome we can see that the numbers don't have a great connection with distance and are not revealing a reliable correlation.

Conclusion

It is apparent that there is no correlation between the distance from the ocean and the concentration of these contaminants. The only data potentially indicating there is one is the linear curve in the PFOS on the box chart, and the low t-test p-value. That said, any observed data on PFOS which isn't seen at all by another contaminant like PFOAs is not likely to be spread in a way that contamination from the ocean. Any correlation seen between concentration and distance from the ocean is likely from differing populations in major inland cities which are being surveyed more frequently. These densely populated areas would spread the contaminants more than remote areas closer to the ocean, and many beach towns aren't very busy for nine months of the year while there are many inland towns more distant from the ocean.